ULTRON 2.7 Phase 8 Execution Report

Voice interaction, transcript history, and speech response layer

Project	ULTRON 2.7
Phase	Phase 8: Voice Mode
Workspace	Local workspace checkout: U2.7
Status	Implemented and verified

Executive Summary

Phase 8 adds voice interaction to ULTRON 2.7. The interface can listen through browser speech recognition when available, send transcripts through the local API, process commands with the existing brain/runtime, and speak responses through browser speech synthesis.

Voice mode remains an adapter around the safe runtime. It does not bypass typed tool calls, schema validation, policy gates, executor restrictions, audit logging, or confirmation requirements.

Phase 8 Scope

- Add a backend voice module with STT/TTS provider abstractions.
- Add voice API endpoints for start, stop, transcribe, speak, and status.
- Add microphone, push-to-talk, mute, stop-speaking, confirmation, and mock voice controls.
- Use browser speech-recognition events plus empty-transcript rejection as the lightweight VAD path.
- Add transcript history covering user speech, understood command, selected tool, result, and spoken response.
- Keep typed commands and subtitles available as fallbacks.
- Add tests with mocked speech providers and safe confirmation behavior.

What Was Implemented

Area	File(s)	Result
Voice runtime	src/ultron27/voice.py	VoiceSession, transcript records, provider protocols, confirmation phrase handling, browser TTS delegation, and mock TTS.
Local API	src/ultron27/web_server.py	Added voice start, stop, transcribe, speak, and status endpoints.

Area	File(s)	Result
Web interface	web/index.html, web/styles.css, web/app.js	Added voice controls, transcript history, browser speech recognition, browser speech synthesis, mute, and stop voice.
Launcher	scripts/run_phase8_voice_ui.py	Adds a Phase 8-named launcher for the same local interface server.
Smoke check	scripts/phase8_voice_smoke.mjs	Adds browser smoke coverage when Playwright's browser binary is available.
Docs	README.md, docs/architecture.md, docs/phase8_voice_mode.md	Documents voice flow, endpoints, safety boundary, and run instructions.
Tests	tests/test_pipeline.py	Expanded to 42 tests including mocked STT/TTS and high-risk voice confirmation.

Voice Pipeline

Stage	Purpose	Safety Property
Capture	Browser speech recognition or mock voice provides text.	Capture does not execute actions; empty speech is ignored.
Transcription API	POST /api/voice/transcribe accepts transcript payloads.	Wake word is stripped and empty speech is rejected.
Brain/runtime	Transcript is processed as a normal goal.	Planner, validation, policy, executor, and audit remain authoritative.
Confirmation	High-risk tasks pause for yes confirm or clicked confirmation.	Destructive actions remain dry-run or not implemented.
Speech output	Response text is sent to /api/speak and browser speech synthesis.	Mute and stop-speaking controls remain available.

User Controls

Control	Behavior	Fallback
Mic On/Off	Starts or stops browser speech recognition.	Typed command input remains available.
Push-to-talk	Switches between one-shot and continuous recognition behavior.	Mock Voice can exercise the pipeline without a microphone.
Mute	Prevents audible ULTRON responses.	Subtitles still show the response.
Stop Voice	Cancels browser speech synthesis and marks speaking stopped.	The UI can return to listening or idle.
Confirm	Sends yes confirm for pending high-risk tasks.	Policy still controls whether execution is allowed.

Verification Results

Check	Command	Result
Unit tests	$env:PYTHONPATH='src'; python -m unittest discover -s tests	42 tests passed.
Compile check	python -m py_compile src/ultron27/voice.py src/ultron27/web_server.py scripts/run_phase8_voice_ui.py	Voice module, server, and launcher compiled successfully.
HTTP smoke	local ThreadingHTTPServer endpoint check	Voice controls served; start, transcribe, speak, and history endpoints worked.
Dataset evaluation	python scripts/evaluate_dataset.py --split test	Intent 1.0, tool 1.0, argument exact match 0.9614, confirmation 1.0.
Safety regression	python scripts/evaluate_safety_regression.py	362 cases, tool accuracy 1.0, policy action accuracy 1.0.
Browser smoke	node scripts/phase8_voice_smoke.mjs	Clean fallback when the Playwright browser binary is unavailable in the sandbox.

Recommended Next Step

- Plug a local STT provider such as faster-whisper or whisper.cpp into the VoiceSession provider interface.
- Add a local TTS provider such as Piper or pyttsx3 for fully offline voice identity.
- Add wake-word and stronger VAD once the microphone path is stable across target machines.